When are taxi rides most in demand and how does the time and location of the trips affect the total amount charged?

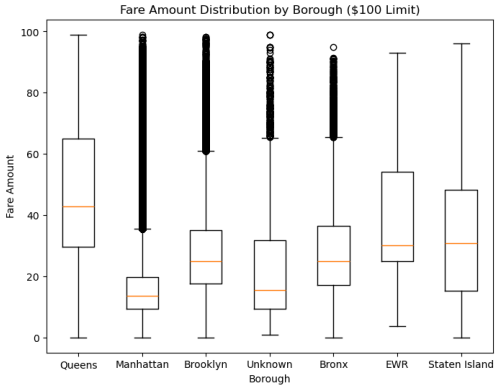
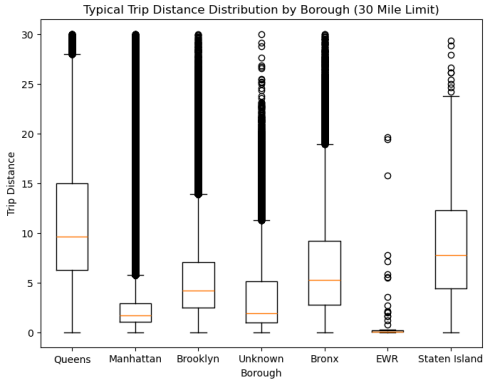
NYC Yellow Taxi Dataset:

The Yellow Taxi Dataset has information from all yellow taxi trips in New York City. With variables such a pickup and dropoff time, pickup and dropoff location ID, passenger count payment type, fare amount, trip distance, one can see a picture of what an average taxi ride looks like in NYC. There are additional variables such as VendorID, RatecodeID, store and forward flag (if there was a server connection), MTA tax, tip amount, tolls amount, improvement surcharge, congestion surcharge, airport fee, and CBD congestion relief fee.

We chose to subset June and July 2025, which housed a total of 8,221,923 trips between the two months and 20 variables (1.1GB of data in total!).

Additionally, for the Geospatial component, we used the NYC Taxi Zones GeoJSON file from NYC OpenData. This includes 6 columns: Shape Geometry, Shape Columns, Shape Area, Zone Name, LocationID, and Borough for 263 NYC taxi zones.

Exploratory Visualizations

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<td>Queens has a noticeably higher average fare amount than the other boroughs and only some of the boroughs count the high fares as outliers while others do not.</td>
<td>EWR has a very small average distance compared to the other boroughs, the rest of them are about the same in average distance while Queens and Staten Island seem to have longer rides which may be the reason for their higher fares. EWR is acting strange, it has one of the highest average fares while also having an incredibly small average distance for the trip.</td>
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Data Cleaning and Preprocessing:

1. Find NA values: Passenger count, ratecodeID, store and forward flag, congestion surcharge, and airport fee all had the same number of missing observations.

   Fill the passenger count with the modeVal (1) as that is what a trip is most likely to have.

   Fill Ratecode with the Null/Unknown ID of 99 (according to the data dictionary).

   The store and forward flag was assumed to be 'N' (not a store and forward trip).

   Congestion surcharge and airport fee were filled with zeros.
2. Removing impossible data:

   dropping rows where the dropoff time was less than the pickup time, or where the fare amount or trip distance was less than or equal to zero.
3. Removing observations with payment types 3 (no charge) and 4 (disputed).
4. Added two rows: "month" based on pickup time (June or July) and "borough" using the NYC taxi zone lookup CSV based on pickup location ID.
5. Adjust the data types for our purposes: passenger count and payment type, and RatecodeID were all turned from float to int data types. Store and forward flag, RatecodeID, improvement surcharge, congestion surcharge, cbd congestion fee, airport fee, payment type, and vendor ID were all changed into categorical variables as the order is not significant.

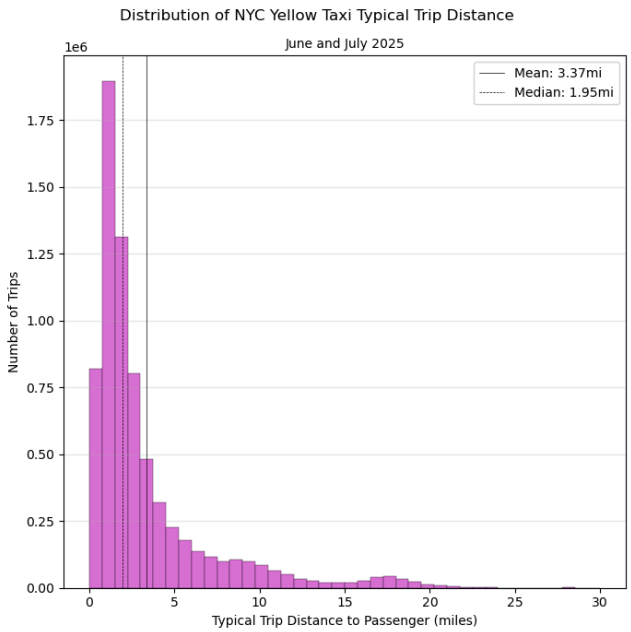
We then took it one step further and made a "typical dataframe" for our visualizations, which consisted of trip distance between 0 and 30 miles and total amount charged between 0 and 100 dollars, as to limit extraordinary outliers that were part of the purpose of our visualizations.

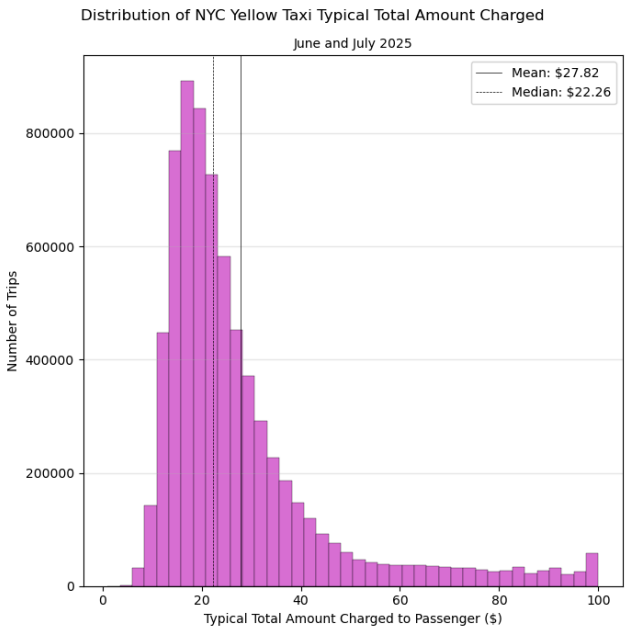
For cleaning the interactive plot:

1. Renaming columns for consistency across datasets.
2. Ensuring the LocationID was an int datatype.

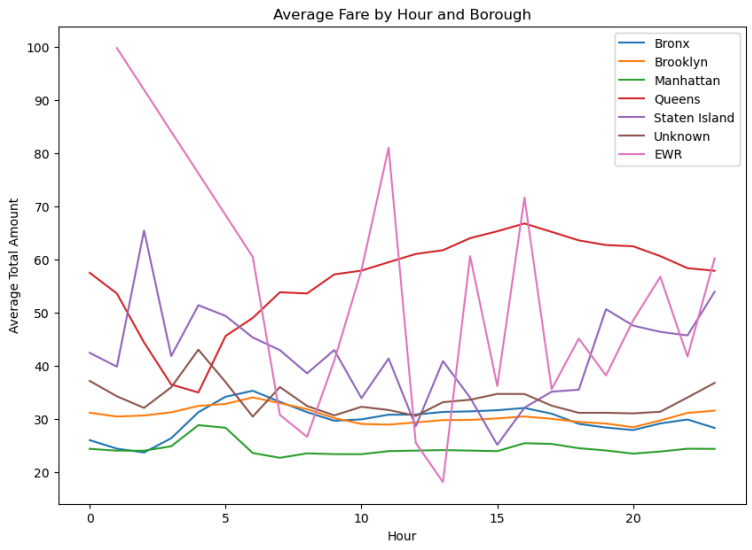
The NYC dataframe was then created by merging an aggregated dataframe of trip_distance and total_distance means grouped by borough with the nyc geo dataset by locationID.

Final Visual Analysis:

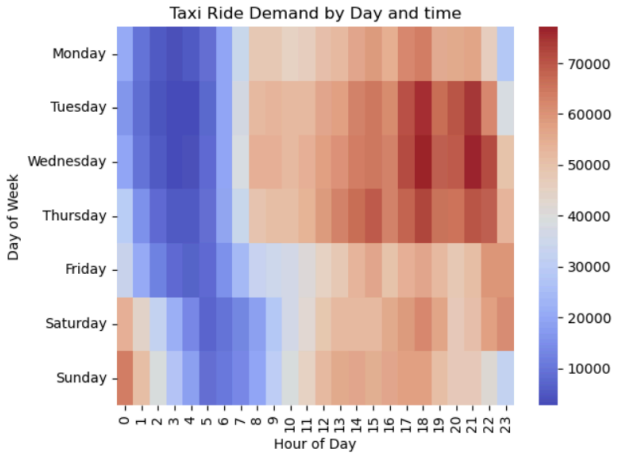
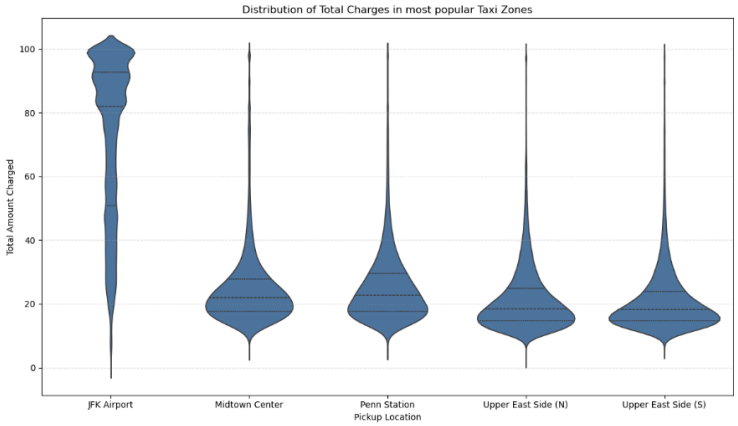




Both distributions are heavily right-skewed, with the majority of trips being less than 5 miles and under $30. It is also important to notice the difference in the means versus the medians. A larger difference means a higher spread of data (more variation). Both distance and total amount appear to have similar spreads, proportionally.



This plot does a good job in showing just how much variation there is between boroughs in regards to average fare amount. EWR has a very high variation, likely due to the greatest variation in distance as it is the furthest borough from the rest. Additionally, Manhattan consistently has the lowest fare amount, likely due the lowest average trip distance. Finally, Queens follows a typical rush hour pattern, with a peak around 3pm and high fares throughout the evening.

Heatmap showing demand for taxi rides by day and time	**Violin plot showing demand in popular taxi zones**
	
The heatmap of taxi demand by day and time shows that there is a noticeable ‘rush hour’ period from about 5-9pm during the weekdays and also a noticeable ‘dead period’ from about 12-7am on the weekdays. Weekends are more consistent throughout the day with a much smaller difference between morning and night as compared to the week days.	The violin plot of the distribution of charges for the 5 locations with the most total rides shows that the popular areas all cost around the same and have similar distributions. That being except for the JFK airport, where total charges are much higher and also are much more spread out. The JFK airport has a much wider range and much higher average price than the other popular areas.

Interactive (HTML in folder):

Working with Google Gemini, we were able to use folium to create an interactive geospatial graph. Starting with the nyc geopandas dataset for Open NYC, we prompted gemini on *“how to make a geopandas dataframe interactive”* and it recommended Folium, which would give us “zoom/pan and basic clickable features.” Then, we prompted it to assist with interactivity and hover to show us important data features, as well as styling.

From there, we were able to add our own field and aliases and personalize the styling to result in an interactive plot that can zoom on click and is colored by borough.

Limitations and Design Choices

Some of the limitations of this dataset was how much it seemed like the airports were the main decider of total amount charged. We had to think of how to work around that to find other ways to show how different factors related to charges. The fact that one of the boroughs is just an airport also skews the data for that borough as all the rides would be airport rides which makes it stand out with noticeably higher charges. That was one of the main reasons that pushed us to look at time of day for our visualizations because that's one of the only ways where airport rides don't skew things too bad due to the fact the airports run at all hours. Another big limitation we had was during the exploratory phase, we saw that EWR has one of the highest average fares while also having an incredibly small average distance of between 1-2 miles. We were unfortunately not able to make a visualization that helped to explain why this was happening which maybe could have been avoided with more complete data.

Conclusion

From our various different visualizations exploring demand for taxi rides and how time and location affects the total charges we were able to discern a few key points. We concluded that time of day does have an effect on the total charges for a taxi but only for some boroughs and not others. Some boroughs will see a significant increase in total charges during rush hours while other boroughs don't see much change. We also concluded that the distance of the trip, and whether or not you are coming or going from an airport are the biggest factors when trying to determine the total charges by taxi rides.